Ground Truth: RSA Correctness from First Principles in Pure Lean 4 Without Mathlib

Davis Brief
davisbrief@gmail.com

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Abstract

We present a complete, machine-checked proof of RSA encryption correctness in the Lean 4 proof assistant, built entirely from first principles without any dependency on the Mathlib library. The development comprises 182 lemmas organized across 12 tiers, beginning with basic natural number arithmetic ($0+n=n$) and culminating in the theorem that RSA decryption inverts encryption ($m^{de} \equiv m \pmod{pq}$ for appropriate primes p, q and exponents d, e). The proof chain passes through divisibility theory, the Euclidean algorithm, prime factorization, Fermat's Little Theorem (proved via binomial coefficients), Bézout's identity, and the Chinese Remainder Theorem. The entire development compiles with zero `sorry` axioms and depends only on Lean's three core axioms (`propext`, `Quot.sound`, `Classical.choice`). To our knowledge, this is the first standalone formalization of RSA correctness in Lean 4 without Mathlib. The proof artifact is publicly verifiable: any user can clone the repository and confirm correctness by running `lake build`.

1 Introduction

RSA [1] is the most widely deployed public-key cryptosystem in the world. Its correctness relies on a chain of number-theoretic results stretching from basic arithmetic through Fermat's Little Theorem and the Chinese Remainder Theorem. While these results are well-known and have been formalized in various proof assistants (notably in Mathlib for Lean 4 [2], and in Coq, Isabelle/HOL, and others), every existing Lean 4 formalization depends on the Mathlib library—a 1.5-million-line dependency that, while powerful, obscures the logical structure of the proof.

We present **Ground Truth**, a self-contained formalization of RSA correctness in 2,553 lines of pure Lean 4 code with zero external dependencies beyond the language's standard library. The development is organized as a linear chain of 12 "tiers," each importing only its predecessor, so that the logical dependency structure is immediately apparent.

Our contributions are:

1. A complete, dependency-free proof of RSA correctness in Lean 4, comprising 182 machine-checked lemmas with zero `sorry`.

2. A proof of Fermat's Little Theorem from scratch, via binomial coefficients and the "freshman's dream" identity, without any finite field or group theory machinery.

3. A constructive proof of Bézout's identity and the existence of modular inverses, using an integer-valued extended GCD.

4. A demonstration that a small LLM (Gemma 4 E4B, 4.5B parameters; DeepSeek-Prover-V2 7B) can contribute meaningfully to formal proof development when paired with Lean's type-checker as an oracle.

2 Architecture

The proof is organized into 12 tiers, each building strictly on the previous:

Tier	Topic	Lemmas	Key Result
1	Natural number basics	20	$a + b = b + a$
2	Ordering	20	Trichotomy
3	Division & modular arithmetic	19	$a = b \cdot \lfloor a/b \rfloor + a \bmod b$
4	Divisibility	19	$d \mid a \wedge d \mid b \Rightarrow d \mid (a + b)$
5	Lists & structural induction	19	$\lvert l_1 \mathbin{++} l_2 \rvert = \lvert l_1 \rvert + \lvert l_2 \rvert$
6	GCD & coprimality	17	Euclid's lemma for coprime pairs
7	Primes	14	$\exists$ infinitely many primes
8	Fermat's Little Theorem	14	$a^p \equiv a \pmod{p}$
9	Bézout & modular inverse	12	$\gcd(a, n) = 1 \Rightarrow \exists b.\ ab \equiv 1 \pmod{n}$
10	Chinese Remainder Theorem	12	CRT existence & uniqueness
11	Fermat-to-RSA bridge	10	$m^{de} \equiv m \pmod{p}$
12	RSA correctness	6	$m^{de} \equiv m \pmod{pq}$
	Total	**182**	

Table 1: Proof chain from basic arithmetic to RSA correctness.

Each tier is a single Lean 4 file. Every lemma carries a plain-English comment explaining what it proves, making the development accessible to non-experts.

3 Key Proofs

3.1 Fermat's Little Theorem (Tier 8)

The proof of $a^p \equiv a \pmod{p}$ for prime p proceeds in three stages:

Stage 1: Binomial coefficient infrastructure. We define binomial coefficients via Pascal's triangle:

$$C(n, 0) = 1, \quad C(0, k + 1) = 0, \quad C(n + 1, k + 1) = C(n, k) + C(n, k + 1)$$

and prove the product identity $C(n, k) \cdot k! \cdot (n - k)! = n!$ by induction on n, with a case split on whether $k < n$ or $k = n$.

Stage 2: Prime divides middle binomial coefficients. Using the product identity and the fact that p does not divide $k!$ or $(p - k)!$ for $0 < k < p$ (proved by showing a prime cannot divide a factorial of smaller numbers), we establish:

$$0 < k < p \implies p \mid C(p, k)$$

Stage 3: Freshman's dream and induction. We define a partial binomial sum $\text{binomSum}(a, n, j) = \sum_{i=0}^{j-1} C(n,i) \cdot a^i$, prove the binomial theorem $(a+1)^n = \text{binomSum}(a, n, n)$, and show that all middle terms vanish mod p. This yields the "freshman's dream":

$$(a + 1)^p \equiv a^p + 1 \pmod p$$

Fermat's Little Theorem then follows by induction on a.

3.2 Chinese Remainder Theorem (Tier 10)

The CRT proof is constructive: given coprime m, n and target remainders r_1, r_2, we compute modular inverses of n mod m and m mod n (via Bézout's identity from Tier 9), then construct the explicit solution $x = r_1 \cdot n \cdot n^{-1} + r_2 \cdot m \cdot m^{-1} \pmod{mn}$.

Uniqueness follows from the key lemma: if $a \equiv b \pmod m$ and $a \equiv b \pmod n$ with $\gcd(m, n) = 1$, then $mn \mid (a - b)$, hence $a \equiv b \pmod{mn}$.

3.3 RSA Correctness (Tier 12)

The capstone theorem states: for distinct primes p, q, if $de \equiv 1 \pmod{(p-1)(q-1)}$, then

$$m^{de} \equiv m \pmod{pq}$$

The proof proceeds in three steps:

1. **Mod** p: Case split on whether $p \mid m$. If yes, both sides are $0 \bmod p$. If no, apply the multiplicative form of Fermat's Little Theorem: $m^{de} = m \cdot m^{k(p-1)(q-1)} \equiv m \cdot 1 \equiv m \pmod p$.
2. **Mod** q: Symmetric argument.
3. **Combine via CRT**: Since $\gcd(p, q) = 1$ (distinct primes are coprime), the Chinese Remainder Theorem yields $m^{de} \equiv m \pmod{pq}$.

4 Proof Methodology

The development was produced through a human-AI collaboration over a single session. The workflow:

1. **Scaffolding**: All 182 theorem statements were written by hand with `sorry` placeholders and verified to compile.
2. **Automated tactics**: A Python orchestrator (`proof_farm.py`) tested each lemma against a battery of 40+ tactic combinations (`omega`, `simp`, `induction`, `cases`, etc.). This solved approximately 60% of lemmas instantly.
3. **LLM-assisted proving**: Lemmas that resisted automated tactics were sent to Gemma 4 E4B (4.5B parameters) and DeepSeek-Prover-V2 (7B parameters) running on a local RTX 2060. Lean's type-checker served as the oracle—candidates were accepted only if they compiled.
4. **Human intervention**: Hard lemmas (Fermat's Little Theorem, Bézout's identity, CRT, prime existence) were proved by the human author with AI assistance for tactic suggestions and arithmetic rearrangements.

DeepSeek-Prover-V2 proved several lemmas on first attempt (e.g., `mod_lt` via `exact Nat.mod_lt _ hb`), while Gemma 4 E4B contributed primarily through trial-and-error on simple arithmetic goals. Neither model could handle proofs requiring novel multi-step reasoning about custom definitions.

5 Axioms and Trust Base

The development depends on exactly three axioms, all part of Lean 4's core:

- `propext`: Propositional extensionality
- `Quot.sound`: Quotient soundness
- `Classical.choice`: The axiom of choice (used via `by_cases` for decidability)

No custom axioms are introduced. The `sorry` count across all 12 tier files is exactly zero.

6 Related Work

RSA correctness has been formalized in several proof assistants:

- **Coq**: Bertot and Castéran [3] include RSA as an exercise in *Interactive Theorem Proving and Program Development*.
- **Isabelle/HOL**: The AFP contains formalizations of RSA-related number theory.
- **Lean 4 + Mathlib**: Mathlib's `ZMod` module provides the infrastructure for RSA, but no standalone proof exists.

To our knowledge, Ground Truth is the first formalization of RSA correctness in Lean 4 that does not depend on Mathlib or any external library.

7 Conclusion

We have demonstrated that a complete, machine-checked proof of RSA correctness can be built from first principles in 2,553 lines of pure Lean 4, organized as a readable 12-tier chain from $0 + n = n$ to $\text{decrypt}(\text{encrypt}(m)) = m$. The development is self-contained, dependency-free, and publicly verifiable.

The proof artifact is available at: `[repository URL upon publication]`

References

[1] R. Rivest, A. Shamir, and L. Adleman. A method for obtaining digital signatures and public-key cryptosystems. *Communications of the ACM*, 21(2):120–126, 1978.

[2] The mathlib Community. The Lean mathematical library. In *CPP 2020*, pages 367–381, 2020.

[3] Y. Bertot and P. Castéran. *Interactive Theorem Proving and Program Development: Coq'Art.* Springer, 2004.